

StructLens

Automated Structural Crack Detection & Analysis System

1. Introduction

Concrete structures — bridges, pillars, dams, and buildings — are cornerstones of modern infrastructure. Over time they develop cracks due to material fatigue, environmental stress, overloading, and natural disasters. Early detection is critical to prevent catastrophic failure, protect public safety, and reduce maintenance costs. Traditional manual inspection is slow, subjective, and prone to missing minor yet critical cracks, creating an urgent need for automated, accurate detection systems.

StructLens addresses this gap by combining OpenCV-based image processing, a Random Forest machine learning classifier, and an integrated Large Language Model (LLM) to automatically detect cracks, quantify severity, and generate professional engineering reports. A Streamlit web interface makes the tool accessible to civil engineers without any programming knowledge.

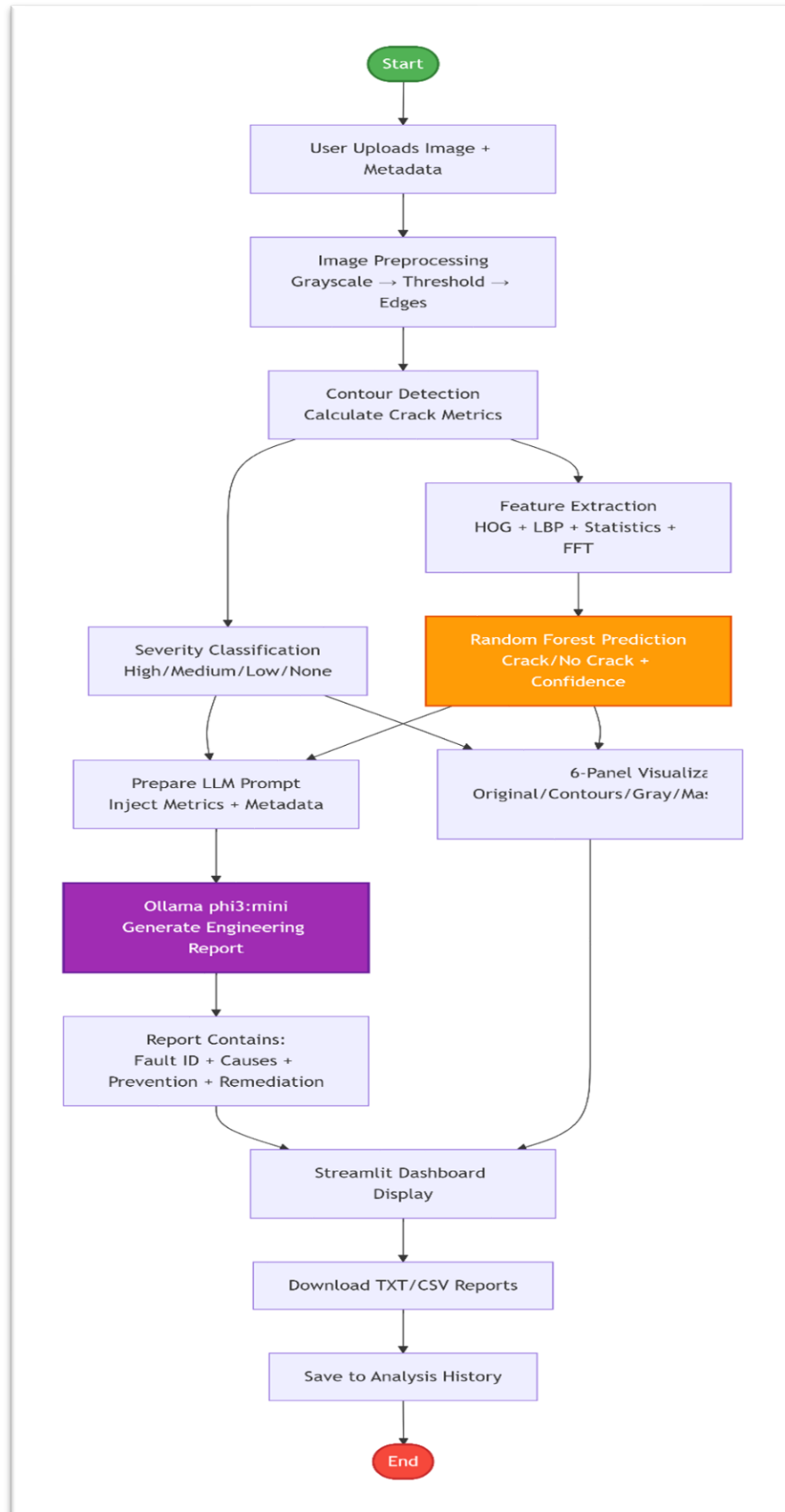
2. Problem Statement

Existing inspection workflows are fragmented: basic computer-vision tools detect cracks but offer no interpretation, while generic report templates lack grounding in actual visual data. No integrated system currently combines precise image-based crack detection with intelligent, context-aware engineering recommendations. Manual reports are inconsistent, and engineers often rely on guesswork — a risk that has contributed to infrastructure collapses and billions in avoidable repair costs worldwide.

3. Objectives

- Automatically detect structural cracks from uploaded images using OpenCV (grayscale conversion, thresholding, contour detection, Canny edge analysis).
- Classify crack severity into four levels — High, Medium, Low, or None — based on calculated crack area and length metrics.
- Predict crack presence using a Random Forest classifier trained on the SDNET dataset with HOG, LBP, statistical, and Fourier-transform features.
- Generate professional LLM-powered engineering reports (fault identification, cause analysis, prevention measures, remediation recommendations) using Ollama phi3:mini.
- Provide a user-friendly Streamlit web interface supporting multiple image formats with downloadable TXT/CSV reports.

4. System Design & Implementation



4.2 Processing Pipeline

Each uploaded image flows through the following sequential steps:

- Image Acquisition – User uploads JPG/PNG/WEBP/BMP/TIFF via the Streamlit interface along with optional metadata (location, structure type, environmental conditions).
- Preprocessing – Grayscale conversion, Otsu thresholding, and Canny edge detection (thresholds 50–150) to enhance crack features and reduce noise.
- Crack Detection & Metrics – Contour detection with area filtering (>50 px) computes crack count, total area, average area, and maximum length.
- Severity Classification – High (area >1000 px, immediate action), Medium (200–1000 px, scheduled repair), Low (50–200 px, routine monitoring), or None.
- ML Inference – Random Forest trained on SDNET (80/20 stratified split) predicts crack presence and outputs a 0–100% confidence score.
- LLM Report Generation – Structured prompt (with real crack metrics as context) sent to Ollama phi3:mini; generates fault ID, root-cause analysis, preventive measures, and remediation steps.
- Visualization & Export – 2×3 multi-panel figure (original, contours, grayscale, mask, edges, severity gauge) displayed in-app; downloadable TXT/CSV reports with timestamp-based filenames.

5. Results

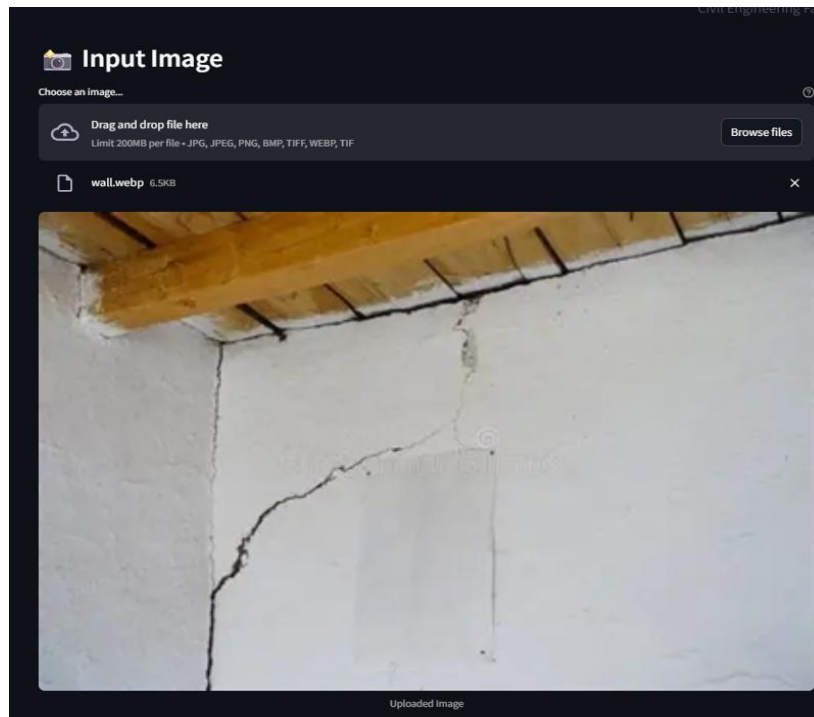


Figure 5.1 Cracked wall



Figure 5.2 Analysis Report

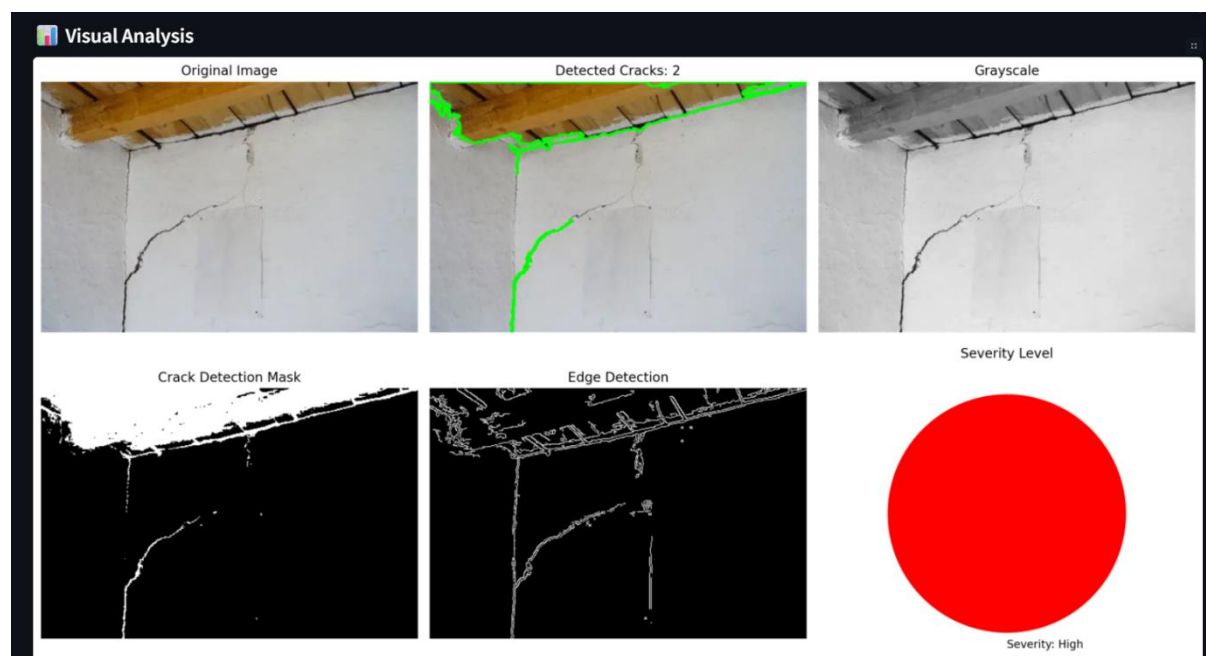


Figure 5.3 Site visual analysis

